Contents

Segmentation and the Ethics of Inferred Categories

> **Supplementary reading — Lesson 8** Estimated reading time: 25 minutes Not examinable. The distinction in Section 2 — that an inferred category has no ground truth to be wrong against — is not mathematics, but it is the kind of thing you should not meet for the first time in a product meeting.

Why a computer scientist should read this

Section 3.3 of the handout ran DBSCAN on Aurora's session data and, as a side effect of finding the bot cluster, produced thirteen sessions the algorithm would not put in either group: **noise**, in the technical sense of "not confidently either." The handout checked their true label — because this is a lesson and the data is synthetic — and reported that all thirteen were genuine humans, customers whose one ordinary week happened to look, by chance, almost as mechanically regular as a script's. That check is a luxury a real bot-detection pipeline never has. Somewhere behind Aurora's thirteen noise points sits a decision — a CAPTCHA (Completely Automated Public Turing test to tell Computers and Humans Apart) challenge, an order hold, a rate limit, or nothing — made about a real person on the strength of a label nobody can confirm.

Section 2 of the handout ran k-means on the same retailer's customers and recovered four segments — "window shoppers," "bargain hunters," "occasional big spenders," "loyal high spenders" — that the handout could grade against `retail_data.py`'s true generating groups (an adjusted Rand index (ARI) of 0.985) precisely because this lesson chose to publish a ground truth that a real marketing team never has. Take that asymmetry out and put a person on the receiving end of the four labels — a loyalty tier, an offer, a credit limit, a price — and the two notebooks stop being a clustering exercise and become a case study in a much older problem: what an organisation is allowed to infer about you, from what, and what happens when the inference is wrong and nobody can say so. This reading is not about clustering being unusually dangerous; it is about what changes, legally and practically, the moment the *purpose* of a method built to find whatever structure is there — with no answer key to consult, ever — shifts from "understand our customers" to "decide something about this one."

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1. Segmentation is one of the oldest deployed uses of this family of methods

Before k-means had a name, retailers were segmenting customers by hand: recency-frequency-monetary (RFM) scoring, drawn straight from a customer's purchase history, has been standard direct-marketing practice since the 1990s and predates any of this lesson's algorithms in industrial use. Loyalty programmes — a supermarket card, an airline mileage tier, a coffee-shop app — exist substantially to generate the transaction history segmentation runs on; the point-of-sale discount is, from the retailer's side, the price of the data. What k-means and its relatives added was scale and automation: instead of an analyst hand-building three or four customer archetypes from intuition, a clustering algorithm finds however many groups the data actually supports, continuously, across millions of customers, with no person looking at any individual row.

The most-told version of what this makes possible is the 2012 case of the American retailer Target, reported at length by Charles Duhigg in the *New York Times*: a statistician built a model — the same family of purchase-pattern analysis this lesson's section 2 uses — that inferred pregnancy from shifts in buying behaviour well before a customer had told anyone, and Target began mailing baby-product coupons accordingly. What made the story travel was not the model's accuracy but a father who complained that his teenage daughter was being sent baby coupons, then called back days later to apologise — she was in fact pregnant and he had not known. Nothing about the story requires the model to have been wrong. It requires only that an inference, made from data given up for an unrelated reason, reached a parent before its subject chose to share it.

The same shape now runs two markets beyond retail marketing. **Credit and insurance underwriting** cluster applicants or policyholders into risk tiers from behavioural signal — a telematics device scoring driving style into a usage-based insurance premium — and a tier, unlike a marketing segment, routes directly to a price. **Ad targeting** builds "lookalike" audiences by clustering users who behave like an advertiser's existing customers, at a scale — hundreds of millions of people, refreshed continuously — that makes any individual customer's presence in a given audience essentially invisible to them. In every one of these, the mechanism is the same one section 2 walked through by hand: assign every point to its nearest representative, update the representative, repeat. What differs is only what happens after the label is assigned, and Sections 3 through 5 are about exactly that difference.

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2. An inferred category has no ground truth to be wrong against

Section 4 of the handout drew a careful line between **internal** metrics — the silhouette score, computable from the clustering and the data alone — and **external** metrics — the adjusted Rand index, computable only because this lesson happens to know `retail_data.py`'s `true_segment` column. Aurora's actual analysts, the handout said plainly, would have the clustering and nothing to check it against.

Sit with what that means once the clustering leaves the notebook. A supervised classifier trained to predict "will this loan default" is wrong in a sense a court, a regulator, or the model's own author can state precisely: the loan defaulted and the model said it would not, or vice versa, and the ground truth arrives eventually. "Window shopper," "bargain hunter," "loyal high spender" are not that kind of claim. They were never predictions of an external fact that will later be confirmed or refuted; they are *names Aurora chose* for regions of a scatter plot that k-means carved up on the strength of two numbers. There is no future event that reveals a customer's "true" segment, because no such thing exists independently of the clustering that produced it. This is not a shortcoming of k-means or of this dataset — it is definitional. Unsupervised methods do not predict a label that already has a meaning; they *manufacture* the label, and the label's only claim to correctness is that it is a self-consistent, reasonably compact grouping of the data it was run on — precisely what an internal metric like the silhouette score checks, and all it checks.

The practical consequence is that "the model is wrong" stops being a question with a computable answer. For an inferred segment, the only things that can be said are weaker and more contestable: the grouping is *unstable* (a different sample of customers, or a different value of k, carves the boundary somewhere else); the grouping is *not useful* for the purpose it is put to (a "low-value" segment mostly containing customers who would spend more given a better offer, which the clustering has no way to know, because it was never told what would have happened under a different offer); or the grouping *causes harm when acted on*, a claim about consequences, not correctness. All three are judgements a person has to make, informed by the data but not settled by it — exactly what section 4's closing point about internal metrics warned against treating as a general endorsement: a clustering can be self-consistent and still be the wrong thing to build a decision on top of.

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3. Proxy discrimination: recovering a protected characteristic nobody asked for

Aurora's clustering pipeline, as built in this lesson, never sees a customer's race, sex, disability status, religion, or any of the other categories that anti-discrimination law in most jurisdictions singles out for protection. Section 2's k-means sees two numbers — annual spend and visit frequency — and section 3's DBSCAN sees two more — session duration and pages viewed. Neither algorithm has any mechanism for using a protected characteristic, because neither is given one.

That is not the same as the resulting clusters being independent of those characteristics, and the gap between the two is what **proxy discrimination** names. Spend and visit frequency, or their real-world analogues — postcode, device type, time of day, which product categories a basket contains — routinely correlate with protected characteristics for reasons that have nothing to do with the algorithm: residential segregation makes postcode a strong proxy for race in many countries; short, interrupted browsing sessions can correlate with disability or caregiving; certain product categories correlate with gender or pregnancy, as the Target story shows directly. A clustering algorithm handed only "innocuous" features can reconstruct, as an emergent property of the geometry it finds, a grouping that lines up closely with a protected characteristic — not because it was told to, but because that characteristic was one of the real causes of the pattern in the innocuous features. Section 5's own eight-column account table makes the mechanism vivid even though its columns are not sensitive: three latent factors — spending propensity, engagement, price sensitivity — generate all eight observed columns, so knowing the eight lets principal component analysis (PCA) recover the three almost exactly. Nothing stops a latent factor behind a real customer table from being, in part, a protected characteristic never collected as such.

This is a genuine bind, not just a risk to be careful about. Checking whether a clustering has proxy-discriminated — the unsupervised analogue of lesson 4's confusion-matrix-per-group check — requires knowing group membership on the very characteristic the clustering was built without. Under European Union (EU) law (Section 5), that characteristic is usually **special category data** — the term used by the General Data Protection Regulation (GDPR), the EU's data protection law, for data revealing racial or ethnic origin, political opinion, religion, trade union membership, genetic or biometric data, health, or sexual orientation — restricted by default, with only narrow exceptions. Auditing a segmentation for exactly this failure may need collecting the one category of data an organisation is otherwise discouraged, or in some cases not permitted, to hold. There is no tidy resolution offered here; Barocas and Selbst's paper in the further reading works the tension through properly.

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4. Two individual harms this lesson's own notebooks produced

Two concrete failure modes are already sitting in the handout, not invented for this reading.

Being the wrong kind of ambiguous. Section 3.3's thirteen noise points are DBSCAN's honest admission that it cannot confidently place those sessions in either the bot or the human cluster — the handout calls this "a more honest output than a forced binary label would be," and it is, *as an output*. What it becomes downstream is a separate question the handout never has to answer, because the lesson stops at the notebook. A rate limiter, a step-up authentication challenge, an account hold, a manual review — any of the ordinary things a bot-detection pipeline does with an ambiguous verdict costs the thirteen genuine customers behind it time, friction, or a blocked purchase, for behaving, by chance, a little more consistently than usual for one week. DBSCAN did not misclassify them — noise is a legitimate, well-defined output, distinct from a wrong label. But a system built on top of it has to decide what "not confidently either" *means* for the person on the other end, and "possibly a bot" is a very different consequence from "possibly a very consistent human," even though the algorithm's evidence for both is the identical sparse neighbourhood.

Being the low-value segment. Section 2's four true customer groups include "window shoppers" — 500 of Aurora's 2,000 customers, built with a mean annual spend of €80 and 1.4 visits a month, the smallest spend and lowest engagement of the four segments the notebook recovers. Nothing in the notebook says what Aurora does with that label, because the lesson stops at discovering it — but the obvious commercial use of a low-value segment is to spend less marketing budget on it, offer worse loyalty terms, or deprioritise its service queue, on the entirely reasonable logic that resources are finite. The individual harm is not that the segmentation is inaccurate — by construction here it is nearly perfect, ARI 0.985 — but that being correctly placed in "window shoppers" can become a self-fulfilling floor: worse offers produce less reason to spend more, which confirms the segment, which justifies the next round of worse offers. A supervised model predicting churn at least has a future outcome that can eventually contradict it. A marketing segment, absent a deliberate experiment offering the "wrong" treatment to a control group, has no such correction built in — a direct consequence of Section 2's point: there is no outcome waiting to prove the inferred category wrong.

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5. What the regulation actually says

Three provisions of the GDPR bear directly on Aurora's two notebooks — lesson 7's resource, `07_trees_and_ensembles/Resources/right_to_explanation.md`, covers Article 22 in full and is the place to go for depth beyond this summary.

GDPR Article 4(4) defines **profiling** as any automated processing of personal data used to evaluate personal aspects of a natural person — in particular to analyse or predict economic situation, preferences, interests, behaviour, or location. Section 2's customer segmentation and section 3's bot detection are both, in the regulation's own vocabulary, profiling, whether or not anyone building them thinks of the word as applying to marketing analytics.

Article 21(2) gives a data subject the right to **object, at any time and without justification**, to processing of their data for direct marketing purposes — a category that explicitly includes profiling to the extent it relates to that marketing. Unlike Article 22, which protects only decisions "solely" automated with a legal or similarly significant effect, Article 21(2) has no such conditions and no exceptions available to the controller: once a customer objects, their data must stop being used for marketing, full stop. Section 2's clustering — used, in the handout's own framing, "for a marketing campaign" — sits squarely inside this provision. Article 22 itself addresses a narrower, separately governed situation: a decision based *solely* on automated processing that produces legal effects or similarly significantly affects the person, a rejected credit application being the standard example. Whether an automated fraud hold following a DBSCAN noise verdict counts depends on how automated the downstream decision is and how significant its effect is judged to be — questions lesson 7's resource shows are genuinely disputed even for a squarely covered case like a loan refusal, and no less disputed here.

Recital 71 — non-binding, as lesson 7's resource explains recitals are, but used by courts to interpret the operative articles — says a controller should use appropriate mathematical and statistical procedures for profiling and take measures that prevent discriminatory effects on the basis of the special categories listed in Section 3. It is the closest the regulation comes to naming, directly, the proxy-discrimination problem Section 3 describes — without using that term, and without resolving the tension Section 3 raised, that checking for such an effect usually requires the very data the regulation otherwise restricts. None of this settles what Aurora must do; it fixes the vocabulary a real pipeline has to be checked against.

What to take from it

- **An inferred category is not a prediction of an external fact — it is a name chosen for a pattern the algorithm found**, so "the model is wrong" has no computable meaning the way it does for a supervised label. What can be assessed is whether the grouping is stable, useful, and harmless when acted on — none of which a silhouette score answers.
- **A clustering built entirely on innocuous features can still recover a protected characteristic**, because both can share an underlying cause. Checking for this usually requires collecting the very category of data privacy law restricts by default — a real bind, not a solved problem.
- **DBSCAN's noise bucket and k-means' lowest-value segment are both legitimate, correct algorithm outputs that still cost a real person something once acted on** — friction for the thirteen genuine sessions of section 3.3, worse terms with no natural correction for the window shoppers of section 2. The harm sits in what was built on the output, not in the output being wrong.
- **GDPR gives marketing profiling an unconditional objection right (Article 21(2)) and gives solely automated, significant decisions a narrower, still-disputed set of protections (Article 22)** — which one applies depends on what the segmentation is used to do, not on which algorithm produced it.

Where to look next

Resource	Type	Why read it
Regulation (EU) 2016/679, Articles 4(4), 21, 22 and Recital 71	Primary legal text	The provisions Section 5 summarises, in full, at `eur-lex.europa.eu`
Barocas & Selbst, "Big Data's Disparate Impact," *California Law Review* (2016)	Paper	The proxy-discrimination mechanism in Section 3, and the audit-data bind, argued rigorously
Duhigg, "How Companies Learn Your Secrets," *New York Times Magazine* (2012)	Long-form journalism	The Target pregnancy-prediction case in Section 1, told in full
Lesson 7's supplementary reading, *The Right to an Explanation*	This course	Article 22 and the explanation debate, in the depth Section 5 only summarises
Barocas, Hardt & Narayanan, *Fairness and Machine Learning*, chapter on unsupervised and representation-level fairness	Book	The formal treatment of proxy variables and fairness without ground-truth labels